Lesson 12.3 - Tests of Independence Using Chi-Square Distribution

> **Video:** Test of Independence Using Chi-Square Distribution by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The idea

A **test of independence** checks whether two categorical variables in a **contingency table** are related.

	Formula
H_0	the variables are **independent**
H_a	the variables are **dependent** (related)
Expected count for each cell	$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$
Test statistic	$\chi^2 = \sum \frac{(O-E)^2}{E}$ over **all** cells
Degrees of freedom	$(r-1)(c-1)$, where r = rows and c = columns

It is a **right-tailed** test: reject H_0 if χ^2 exceeds the critical value.

Example - study time vs type of student

Is the number of hours spent studying each day related to whether a student is a freshman or a senior? $\alpha = 0.05$.

Step 1 - complete the observed table

	0-2 h	2-4 h	4-6 h	**Row total**
Freshmen	76	143	91	**310**
Seniors	147	109	64	**320**
Column total	**223**	**252**	**155**	**630**

Step 2 - hypotheses and critical value

- H_0: study hours are **independent** of student type
- H_a: study hours **depend** on student type
- df $= (2-1)(3-1) = 2$, right-tail area 0.05: $\chi^2_{\text{crit}} = 5.99$

Step 3 - expected counts

Example: $E_{1,1} = \frac{310 \times 223}{630} \approx 109.7$

	0-2 h	2-4 h	4-6 h
Freshmen	109.7 (≈110)	124.0	76.3 (≈76)
Seniors	113.3 (≈113)	128.0	78.7 (≈79)

The rounded values still match every row and column total (e.g. 110 + 124 + 76 = 310).

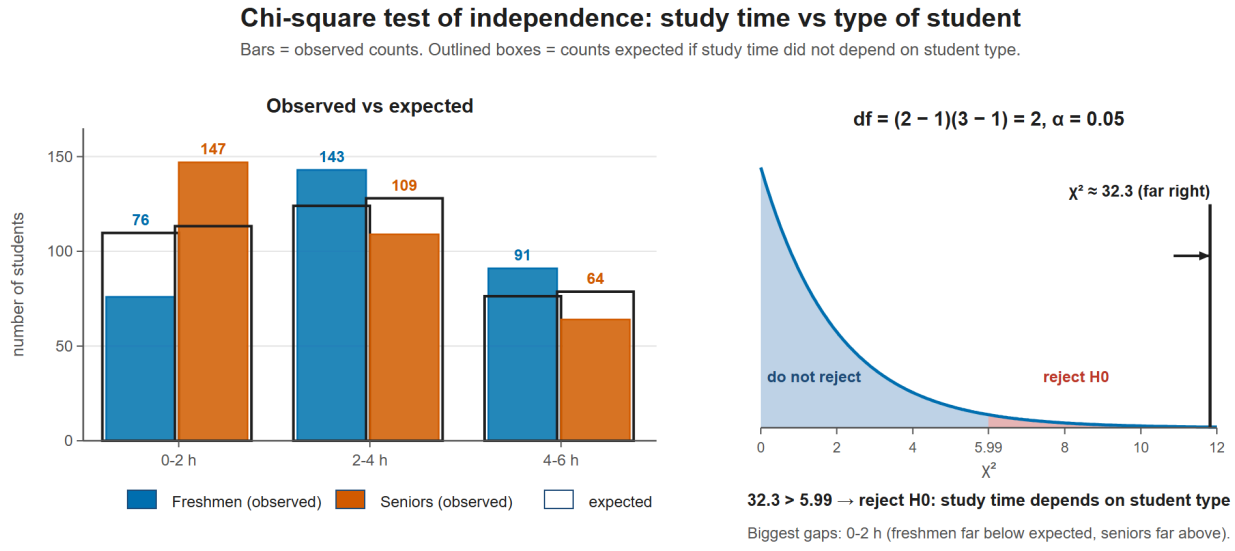
Step 4 - test statistic (with the video's rounded expected counts)

Cell	O	E	$(O-E)^2/E$
Freshmen, 0-2	76	110	10.51
Freshmen, 2-4	143	124	2.91
Freshmen, 4-6	91	76	2.96
Seniors, 0-2	147	113	10.23
Seniors, 2-4	109	128	2.82
Seniors, 4-6	64	79	2.85
Total			**≈ 32.3**

> **Note:** expected counts don't have to be whole numbers. Using the unrounded values (109.73, 124.0, 76.27, 113.27, 128.0, 78.73) gives $\chi^2 \approx 31.7$. The conclusion is the same.

Step 5 - decision

$32.3 > 5.99$, which is in the rejection region. **Reject H_0.** There is strong evidence that study time **depends** on student type. In the data, seniors are much more likely than freshmen to study only 0-2 hours a day.



Left: if study time were independent of student type, the bars would roughly fill the outlined expected boxes. The 0-2 hour group is far off for both freshmen and seniors. Right: $\chi^2 \approx 32.3$ is far beyond 5.99.

Key takeaways

- Expected count = (row total × column total) ÷ grand total.
- $\chi^2 = \sum (O - E)^2 / E$ over every cell, with df $= (r - 1)(c - 1)$.
- Right-tailed: a large χ^2 → reject independence → the variables are related.